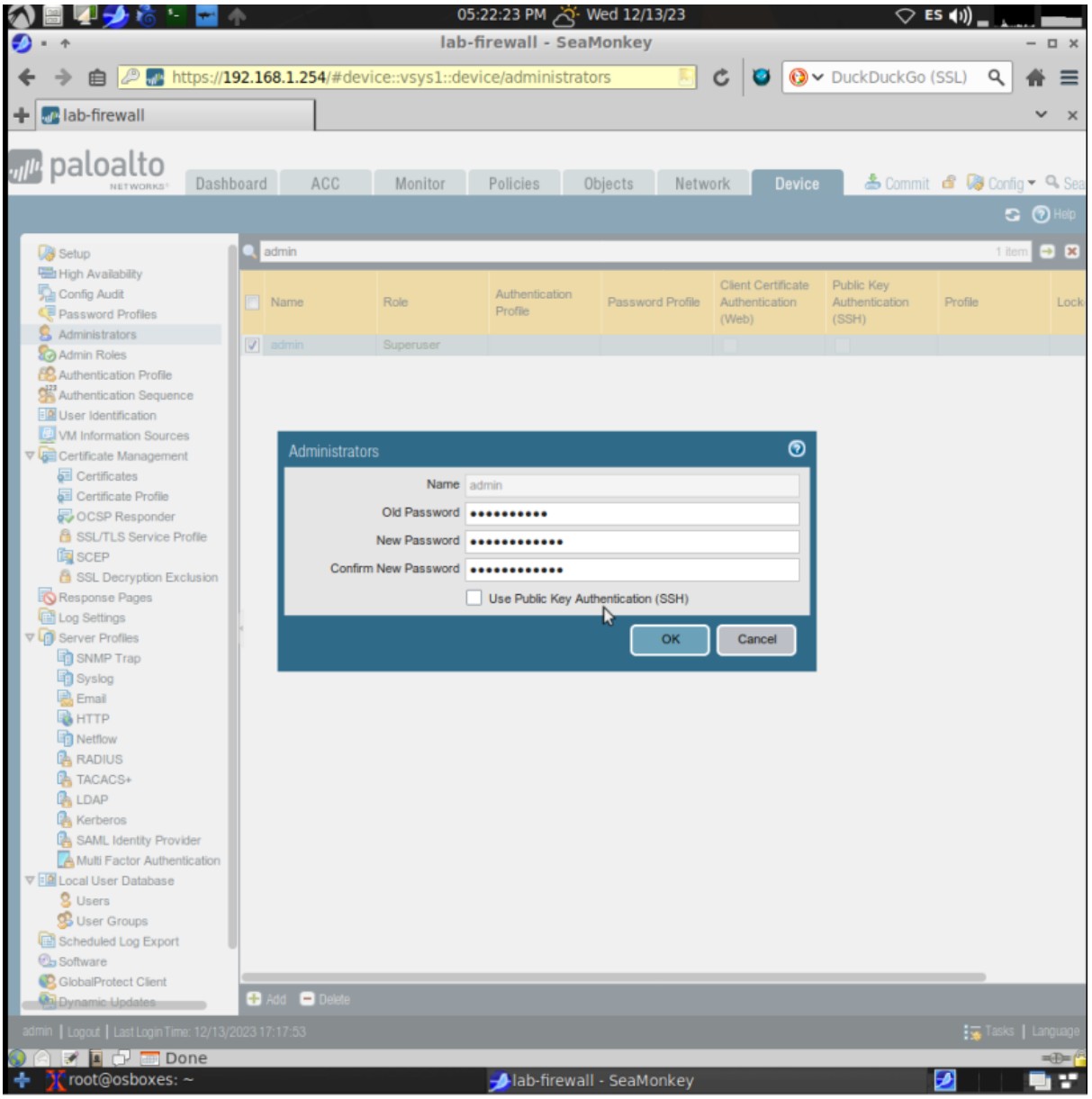
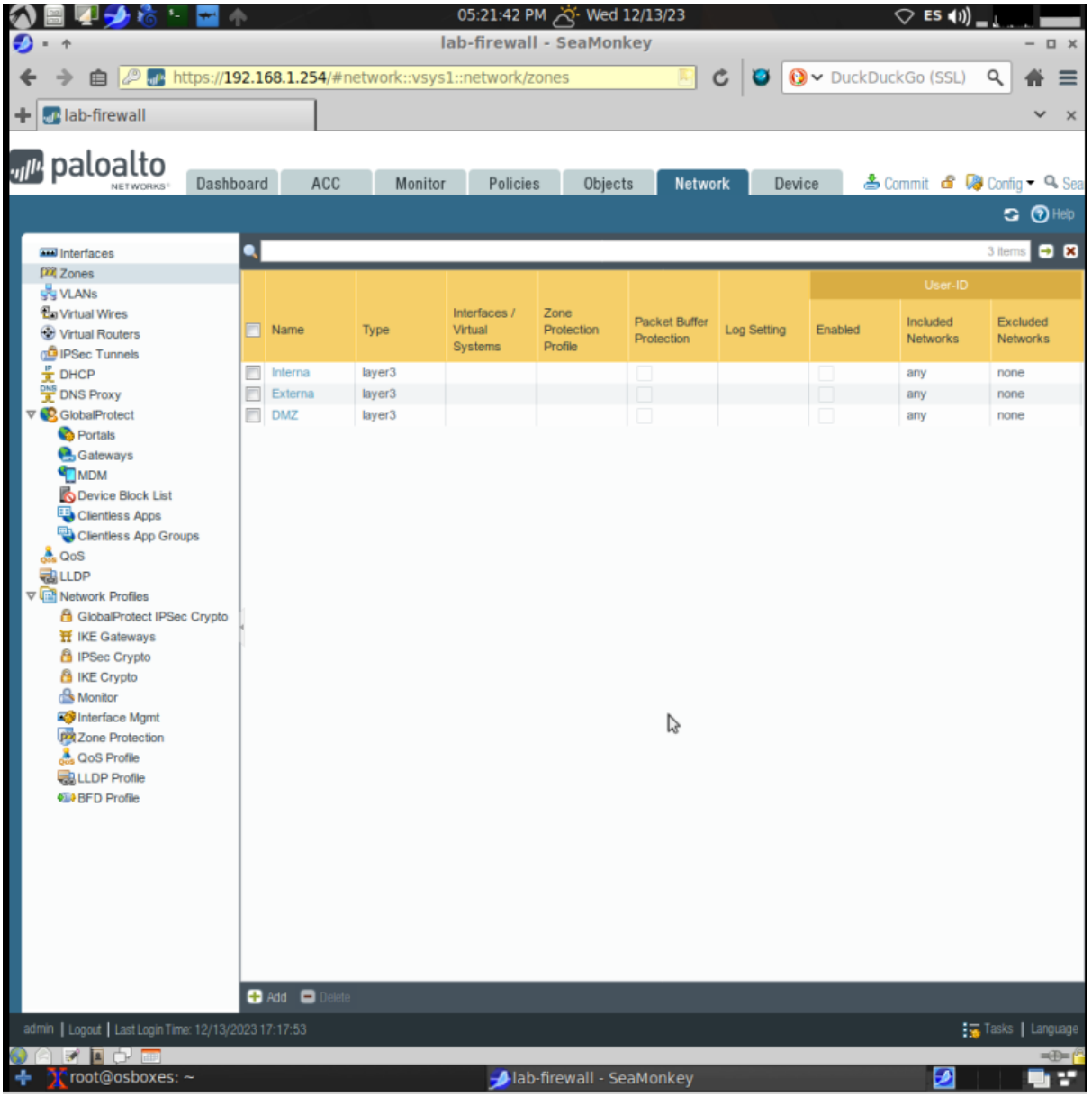
# Memoria Práctica FW

Andrés Roques y Gerardo Zapico

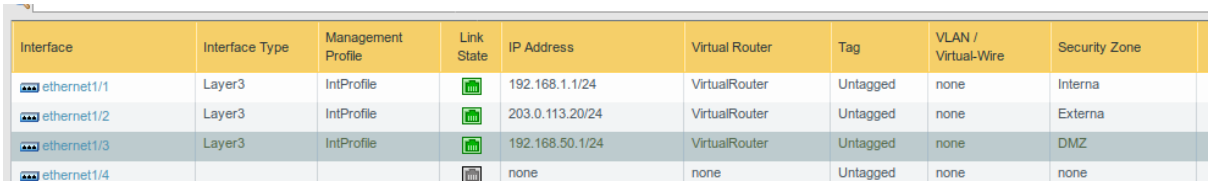
P1


05:22:23 PM Wed 12/13/23
lab-firewall - SeaMonkey
https://192.168.1.254/#device::vsys1::device/administrators
DuckDuckGo (SSL)
lab-firewall
paloalto NETWORKS
Dashboard
ACC
Monitor
Policies
Objects
Network
Device
Commit
Config
Help
Setup
High Availability
Config Audit
Password Profiles
Administrators
Admin Roles
Authentication Profile
Authentication Sequence
User Identification
VM Information Sources
Certificate Management
Certificates
Certificate Profile
OCSP Responder
SSL/TLS Service Profile
SCEP
SSL Decryption Exclusion
Response Pages
Log Settings
Server Profiles
SNMP Trap
Syslog
Email
HTTP
Netflow
RADIUS
TACACS+
LDAP
Kerberos
SAML Identity Provider
Multi Factor Authentication
Local User Database
Users
User Groups
Scheduled Log Export
Software
GlobalProtect Client
Dynamic Updates
admin
1 item
Name
Role
Authentication Profile
Password Profile
Client Certificate Authentication (Web)
Public Key Authentication (SSH)
Profile
admin
Superuser
Administrators
Name admin
Old Password
New Password
Confirm New Password
Use Public Key Authentication (SSH)
OK
Cancel
Add
Delete
admin | Logout | Last Login Time: 12/13/2023 17:17:53
Tasks | Language
Done
root@osboxes: ~
lab-firewall - SeaMonkey

P2


05:21:42 PM Wed 12/13/23
lab-firewall - SeaMonkey
https://192.168.1.254/#network::vsys1::network/zones
DuckDuckGo (SSL)
lab-firewall
paloalto NETWORKS
Dashboard
ACC
Monitor
Policies
Objects
Network
Device
Commit
Config
Help
Interfaces
Zones
VLANs
Virtual Wires
Virtual Routers
IPSec Tunnels
DHCP
DNS Proxy
GlobalProtect
Portals
Gateways
MDM
Device Block List
Clientless Apps
Clientless App Groups
QoS
LLDP
Network Profiles
GlobalProtect IPSec Crypto
IKE Gateways
IPSec Crypto
IKE Crypto
Monitor
Interface Mgmt
Zone Protection
QoS Profile
LLDP Profile
BFD Profile
3 items
User-ID
Name
Type
Interfaces / Virtual Systems
Zone Protection Profile
Packet Buffer Protection
Log Setting
Enabled
Included Networks
Excluded Networks
Interna layer3 any none
Externa layer3 any none
DMZ layer3 any none
Add
Delete
admin | Logout | Last Login Time: 12/13/2023 17:17:53
Tasks | Language
root@osboxes: ~
lab-firewall - SeaMonkey

# P3



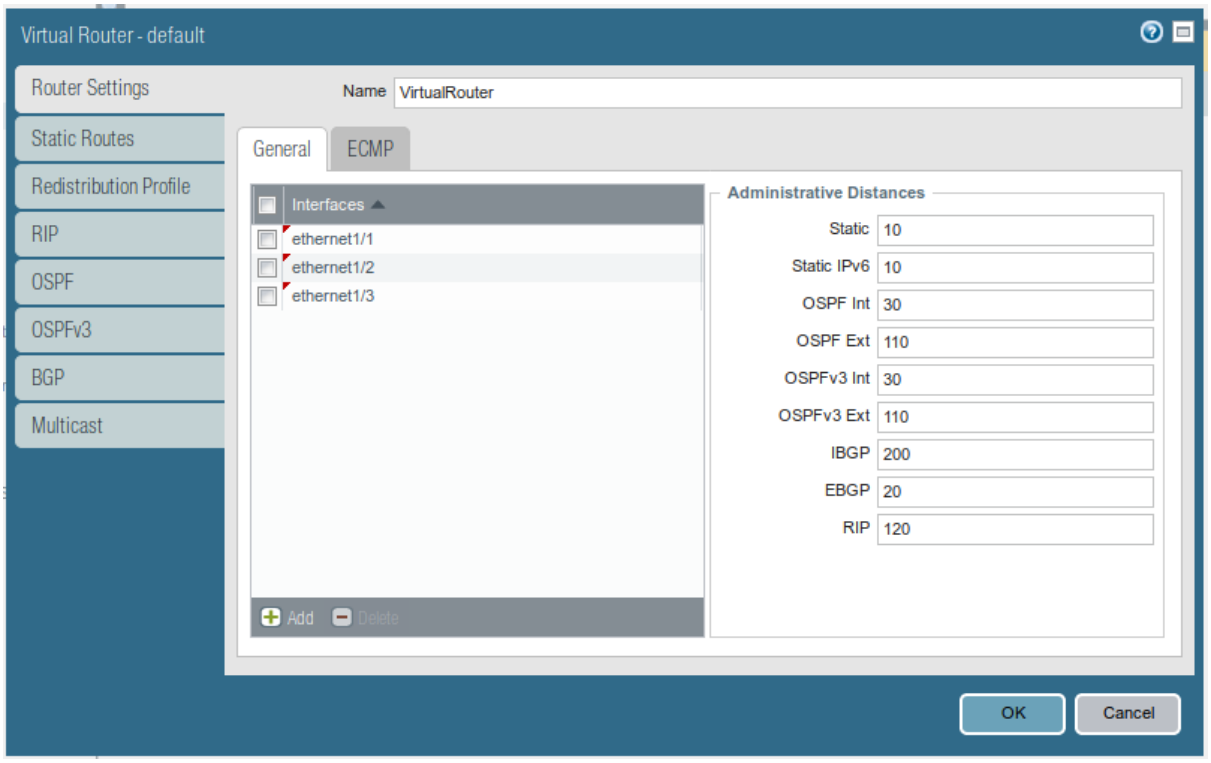
| Interface | Interface Type | Management Profile | Link State | IP Address | Virtual Router | Tag | VLAN / Virtual-Wire | Security Zone |
|---|---|---|---|---|---|---|---|---|
| ethernet1/1 | Layer3 | IntProfile | | 192.168.1.1/24 | VirtualRouter | Untagged | none | Interna |
| ethernet1/2 | Layer3 | IntProfile | | 203.0.113.20/24 | VirtualRouter | Untagged | none | Externa |
| ethernet1/3 | Layer3 | IntProfile | | 192.168.50.1/24 | VirtualRouter | Untagged | none | DMZ |
| ethernet1/4 | | | | none | none | Untagged | none | none |

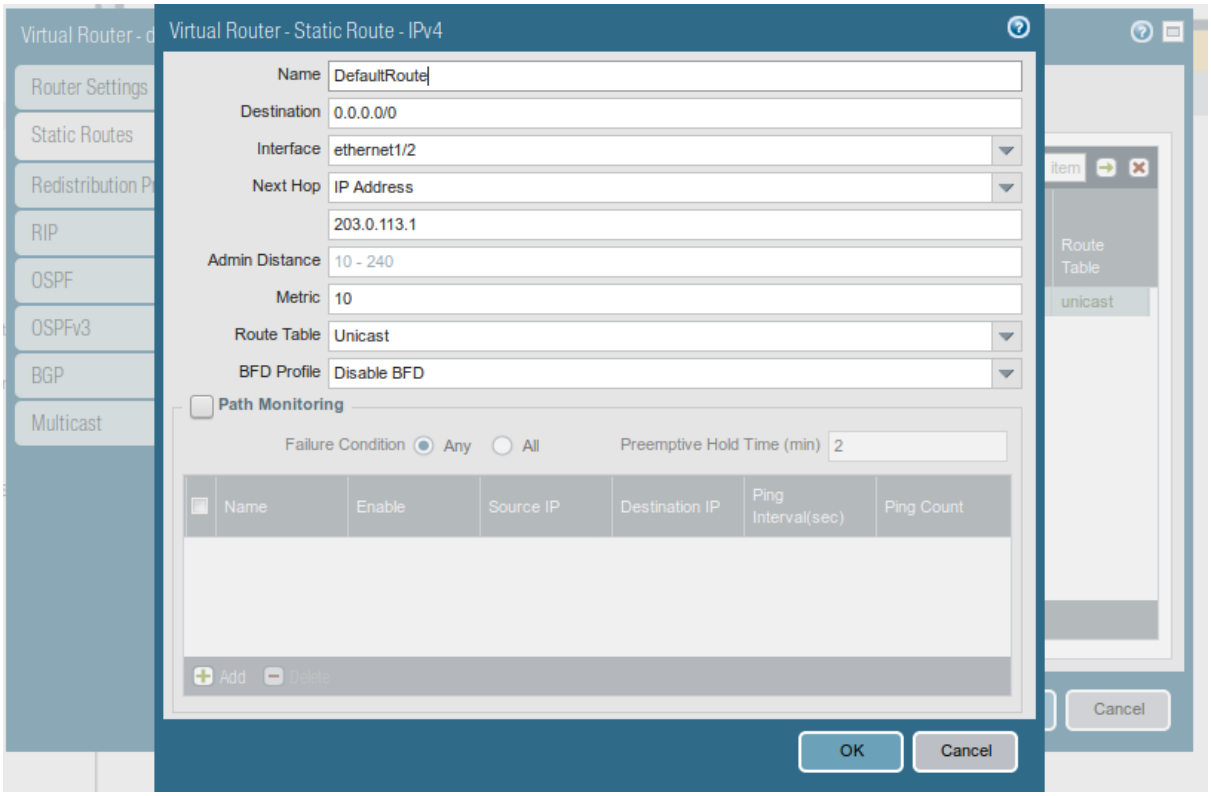
(captura realizada aposteriori, por eso aparecen las demás configuraciones)

**¿A qué red de Virtualbox (vmnetx) corresponde cada interfaz del router (ethernet1/x), a la vista de esta configuración?**

- ethernet 1/1 forma parte de vmnet1
- ethernet 1/2 forma parte de vmnet2
- ethernet 1/3 forma parte de vmnet3

# P4

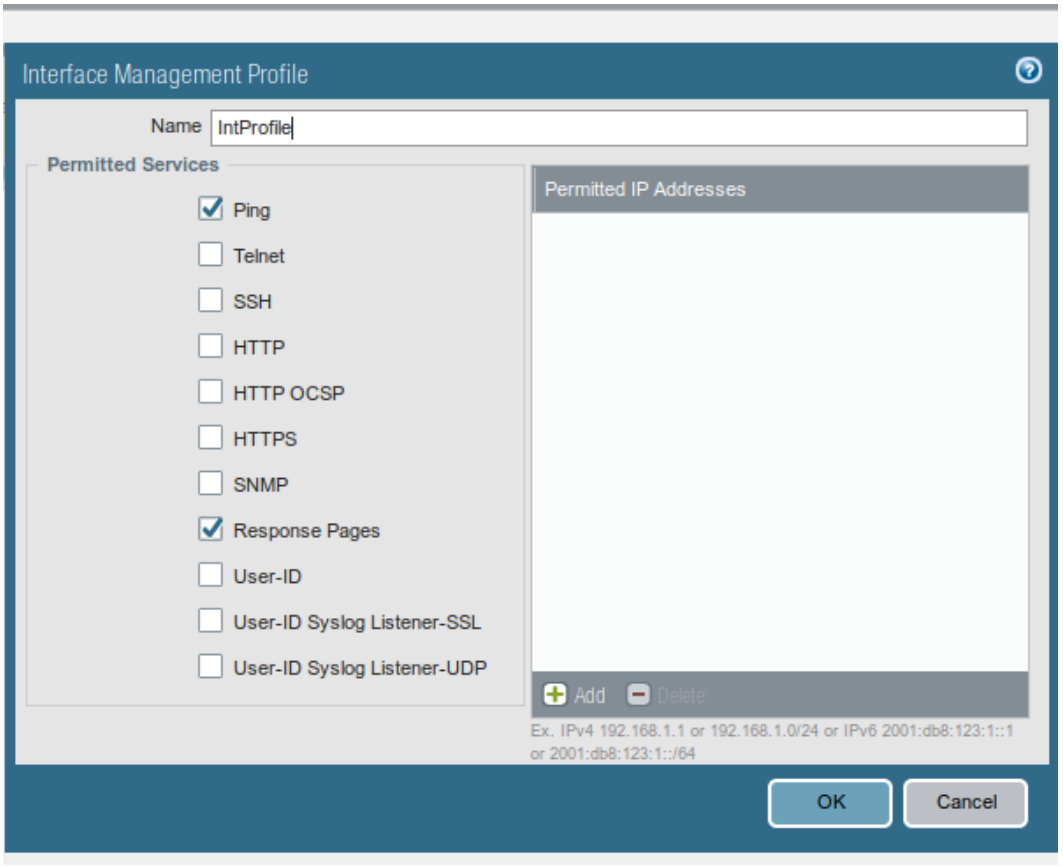

Virtual Router - default
Router Settings
Static Routes
Redistribution Profile
RIP
OSPF
OSPFv3
BGP
Multicast
Name VirtualRouter
General
ECMP
Interfaces
ethernet1/1
ethernet1/2
ethernet1/3
Add
Delete
Administrative Distances
Static 10
Static IPv6 10
OSPF Int 30
OSPF Ext 110
OSPFv3 Int 30
OSPFv3 Ext 110
IBGP 200
EBGP 20
RIP 120
OK
Cancel

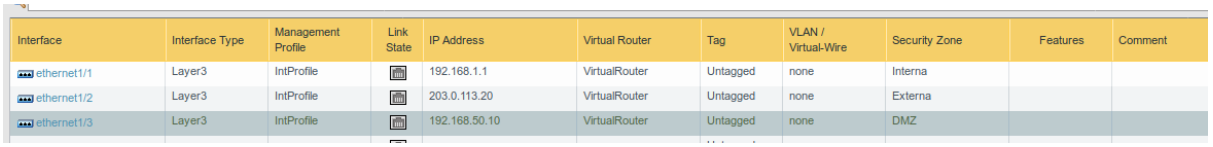

Virtual Router - Static Route - IPv4
Name DefaultRoute
Destination 0.0.0.0/0
Interface ethernet1/2
Next Hop IP Address
203.0.113.1
Admin Distance 10 - 240
Metric 10
Route Table Unicast
BFD Profile Disable BFD
Path Monitoring
Failure Condition Any All
Preemptive Hold Time (min) 2
Name
Enable
Source IP
Destination IP
Ping Interval(sec)
Ping Count
Add
Delete
OK
Cancel

**¿Por qué es necesario añadir esta ruta? Describa para qué tráfico se aplicará y que se hace con ese tráfico?**

Esta será la ruta por defecto (usualmente 0.0.0.0/0) que será útil para el tráfico dirigido a internet. Todos los paquetes dirigidos a una dirección externa a las redes de la empresa que tengan que pasar por internet usará como dirección de destino 0.0.0.0/0.

# P5



| Interface | Interface Type | Management Profile | Link State | IP Address | Virtual Router | Tag | VLAN / Virtual-Wire | Security Zone | Features | Comment |
|---|---|---|---|---|---|---|---|---|---|---|
| ethernet1/1 | Layer3 | IntProfile | | 192.168.1.1 | VirtualRouter | Untagged | none | Interna | | |
| ethernet1/2 | Layer3 | IntProfile | | 203.0.113.20 | VirtualRouter | Untagged | none | Externa | | |
| ethernet1/3 | Layer3 | IntProfile | | 192.168.50.10 | VirtualRouter | Untagged | none | DMZ | | |

# P6

**¿Qué significan las dos reglas de seguridad predefinidas? ¿Es posible el acceso a Internet? ¿Por qué?**

Las dos reglas configuradas predeterminadamente aplican las siguientes políticas:

- intrazone-default: Esta política está configurada de manera que el tráfico dentro de una misma zona pueda concurrir y se permita el paso por el FW. **Action: Allow**

- interzone-default: Esta política está configurada de manera que el tráfico entre zonas no se permita a través del router virtual previamente configurado. **Acion: Deny**

El acceso a internet no es posible, tampoco se recibe el ping en las interfaces del FW. Esto se debe a la regla interzone-default, la cual deshabilita el tráfico entre redes.

# P7

| | Name | Tags | Type | Source | | | | Destination | | Application | Service | Action | Profile | Options |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Zone | Address | User | HIP Profile | Zone | Address | | | | | |
| 1 | universal-default | none | universal | any | any | any | any | any | any | any | any | Allow | none | |
| 2 | intrazone-default | none | intrazone | any | any | any | any | (intrazone) | any | any | any | Allow | none | none |
| 3 | interzone-default | none | interzone | any | any | any | any | any | any | any | any | Deny | none | none |

**¿Es posible acceder a esas interfaces? ¿Puede acceder ahora a Internet? ¿y a otras IPs (por ejemplo, a la del servidor DMZ (192.168.50.10) o la del router VR (203.0.113.1)? Justifique las respuestas.**

Existe conectividad con las interfaces del FW y con el servidor DMZ pero no con el router VR.

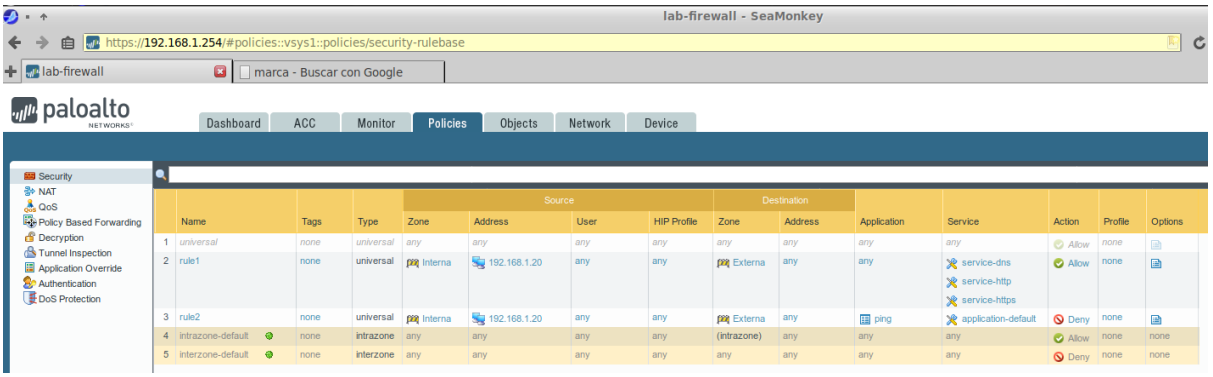
Esto puede deberse a que la dirección de destino del cliente (192.168.1.20) no está correctamente exportada por el router virtual del FW o no está configurado el NAT necesario para que la red externa acepte tráfico de otras subredes. No es bueno que las direcciones IP de la red interna de la empresa sean accesibles desde redes externas y por eso se traducen las direcciones IP.

# P8

**¿Funciona ahora el acceso a otras subredes? ¿A cuáles? ¿Por qué? ¿Y el acceso a Internet? ¿Por qué?**

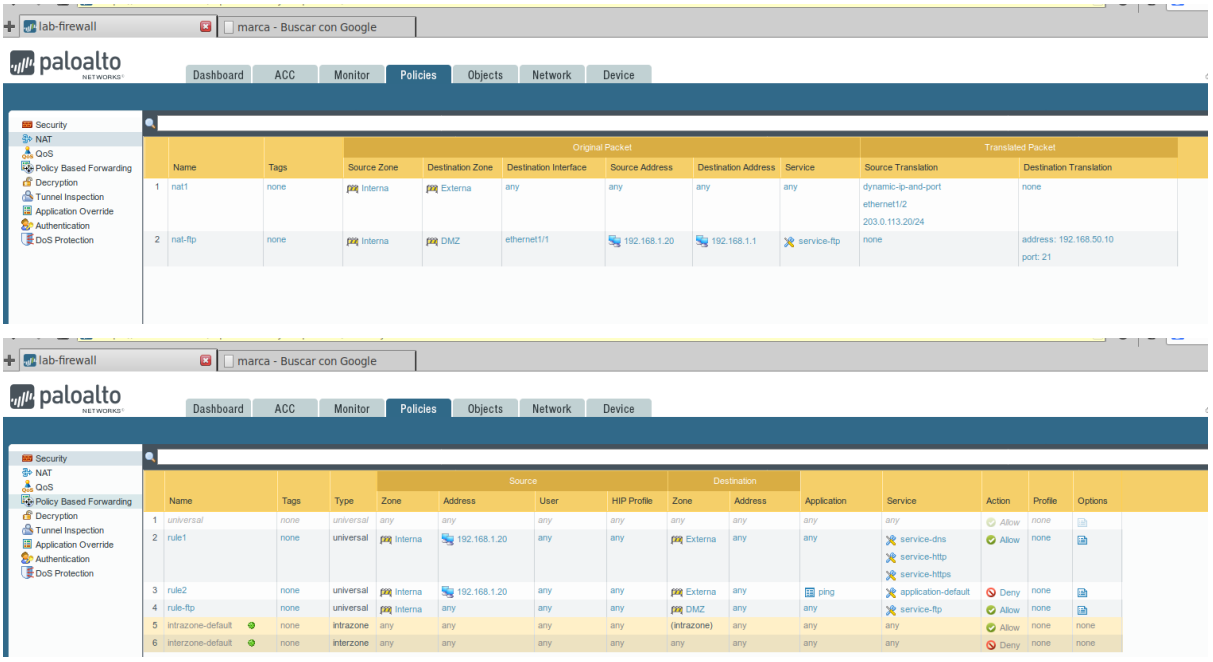
Ahora hay conexión con el servidor DMZ y con el router VR, pertenecientes a otras subredes (zona DMZ y zona Externa). Esto puede conseguirse gracias al uso del Network Address Translation (NAT) que traduce las direcciones de origen de la zona interna por la dirección de la interfaz del router virtual del FW en la zona externa. De la misma manera se consigue acceso a internet a través de la zona externa y el router VR.

# P9



**¿Es posible acceder ahora desde la zona Interna a la zona DMZ? ¿y hacer ping a un destino de Internet?**
Por la norma que bloquea el ping desde la zona interna a la zona externa, los ping a los destinos de internet no podrán llegar a la dirección solicitada, aunque las búsquedas sigan habilitadas (servicios http, https y dns).
No es posible acceder al servidor DMZ.

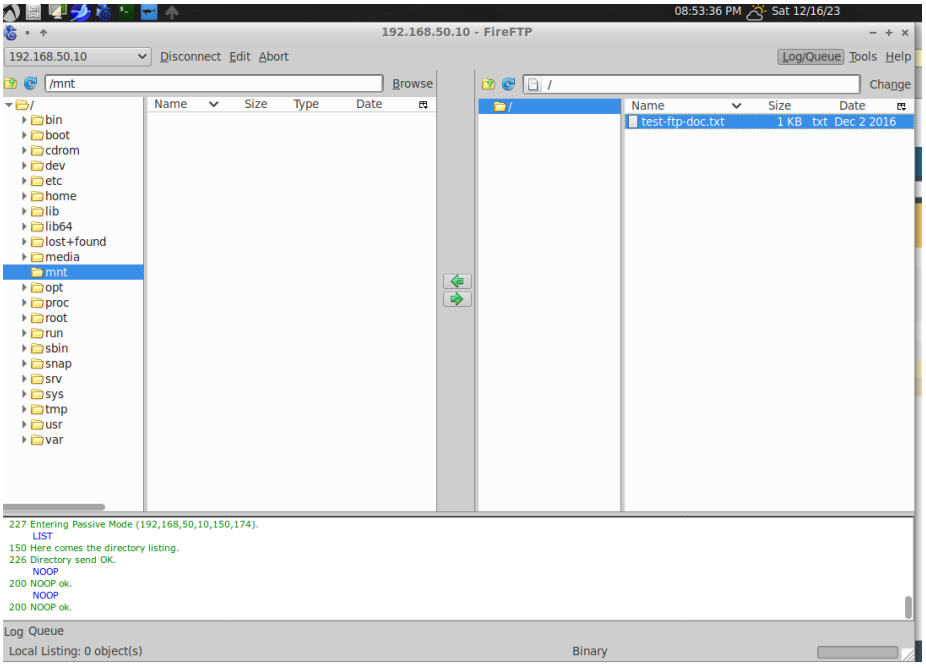
# P10







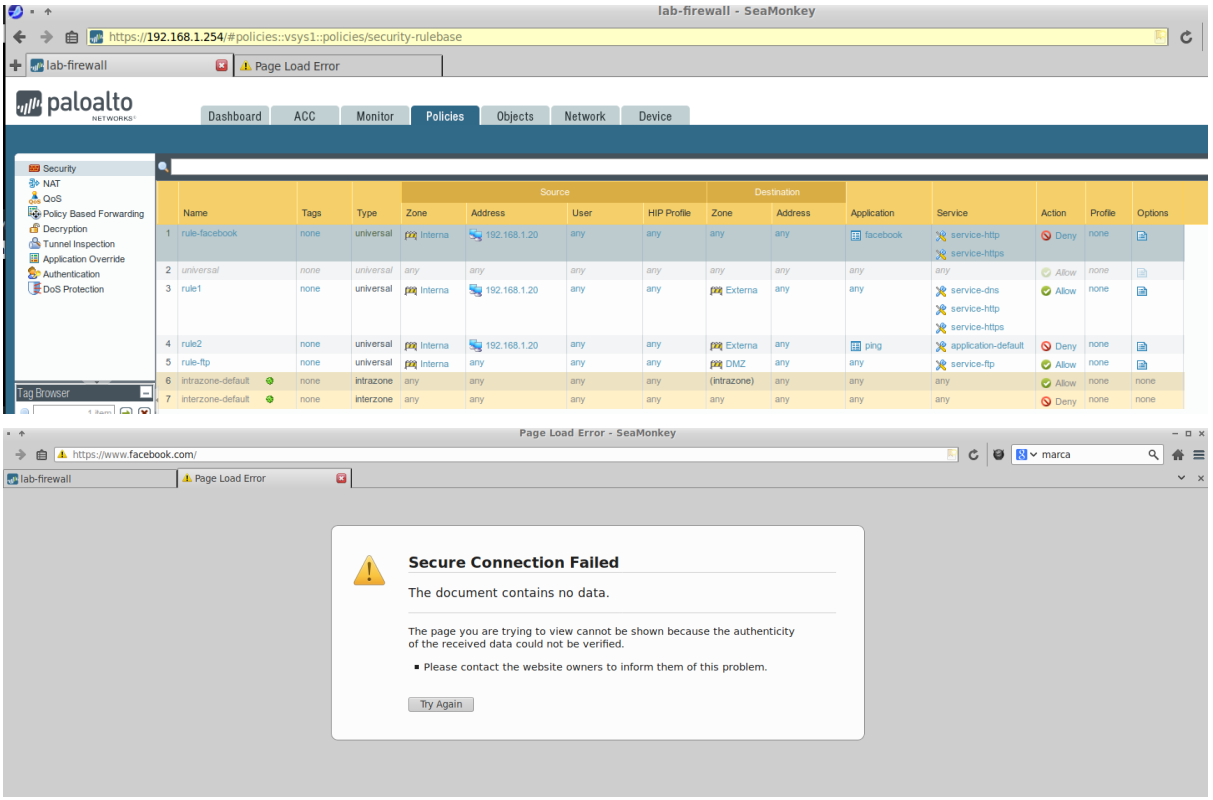
# P11



**Contenido del fichero test-ftp-doc.txt**

test file for ftp access

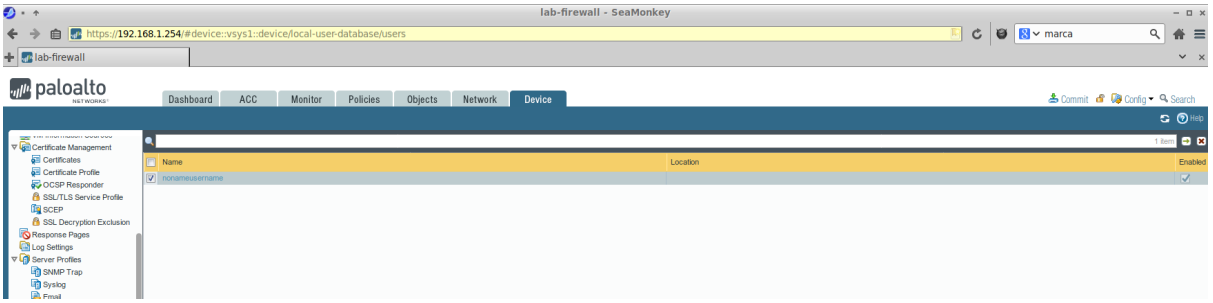
# P12

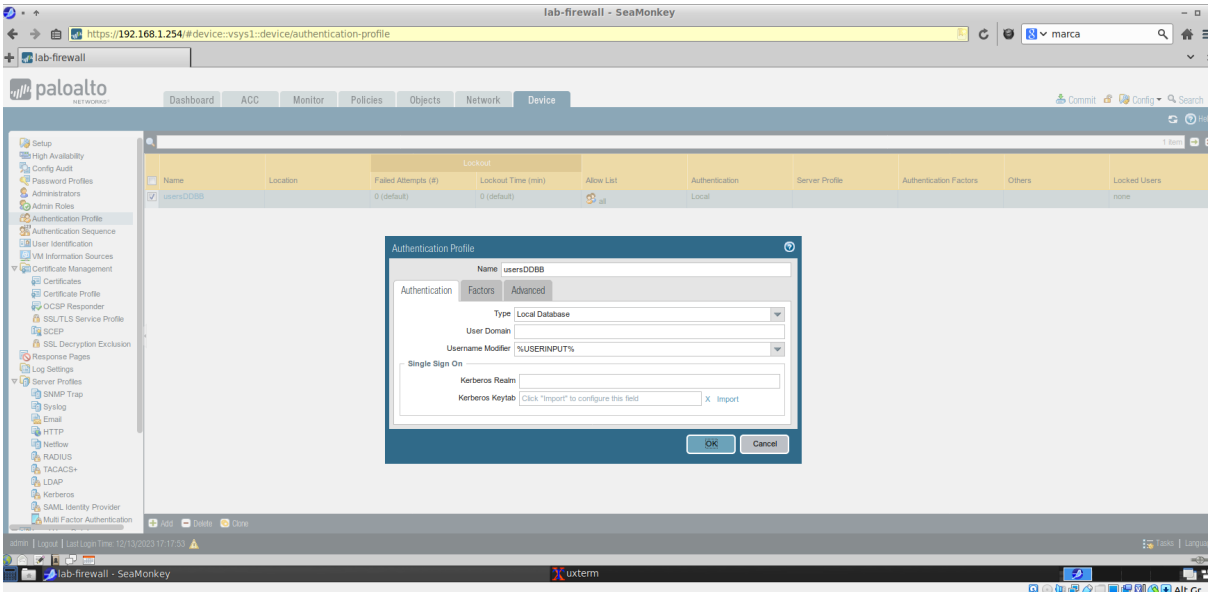


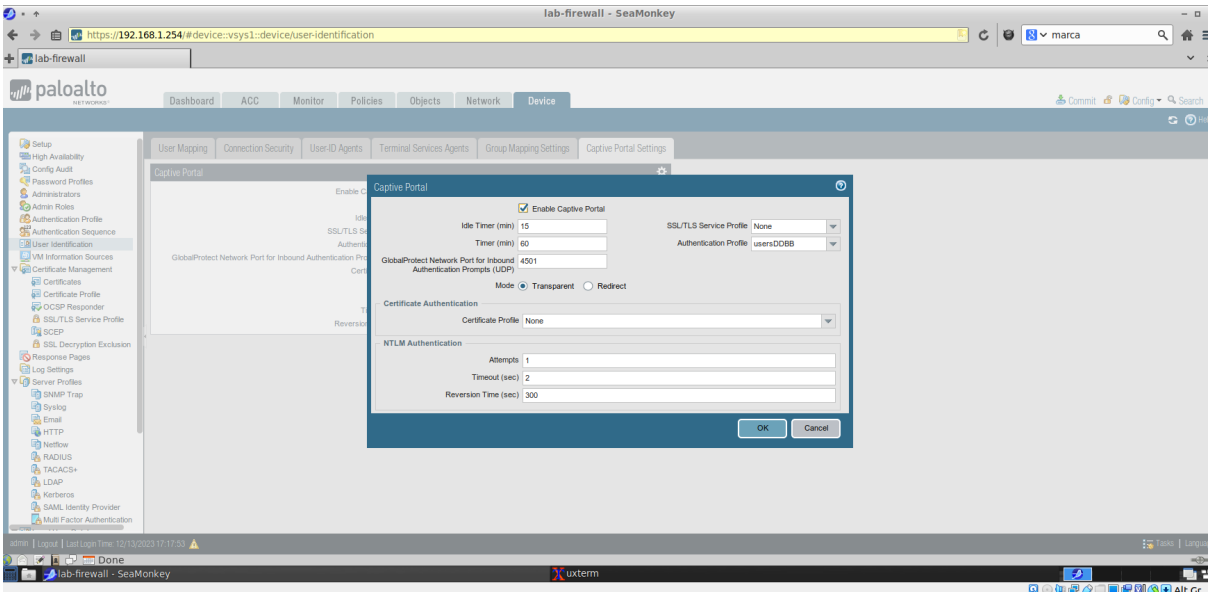

# P13

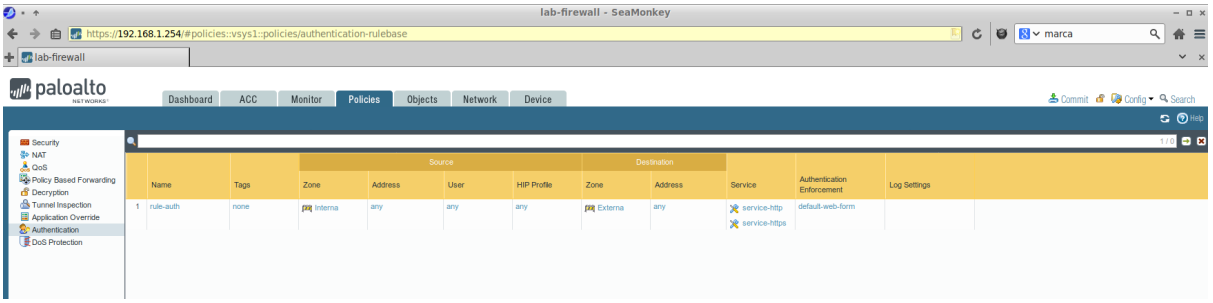


# P14


lab-firewall - SeaMonkey
https://192.168.1.254/#device::vsys1::device/local-user-database/users
marca
lab-firewall
paloalto
Dashboard
ACC
Monitor
Policies
Objects
Network
Device
Commit
Config
Search
Certificate Management
Certificates
Certificate Profile
OCSP Responder
SSL/TLS Service Profile
SCEP
SSL Decryption Exclusion
Response Pages
Log Settings
Server Profiles
SNMP Trap
Syslog
Email
Name
Location
Enabled
nonameusername

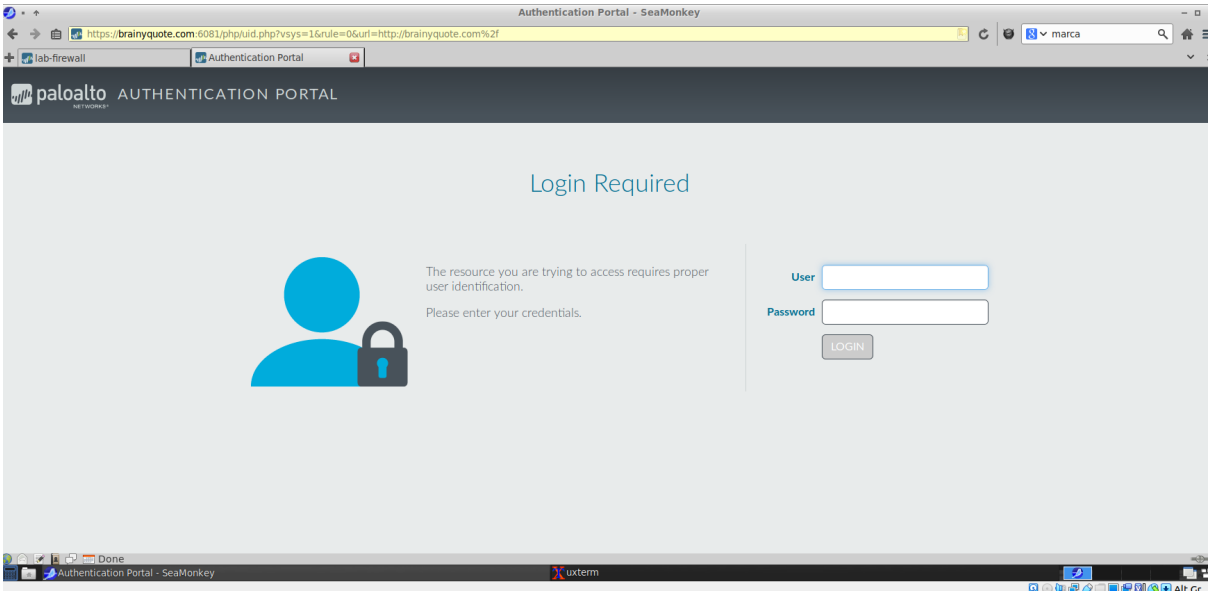

lab-firewall - SeaMonkey
https://192.168.1.254/#device::vsys1::device/authentication-profile
marca
lab-firewall
paloalto
Dashboard
ACC
Monitor
Policies
Objects
Network
Device
Commit
Config
Search
Authentication Profile
Name
usersDDBB
Authentication
Factors
Advanced
Type
Local Database
User Domain
Username Modifier
%USERINPUT%
Single Sign On
Kerberos Realm
Kerberos Keytab
Click "Import" to configure this field
Import
OK
Cancel
Add
Delete
Done
admin
Logout
Tasks
lab-firewall - SeaMonkey
uxterm

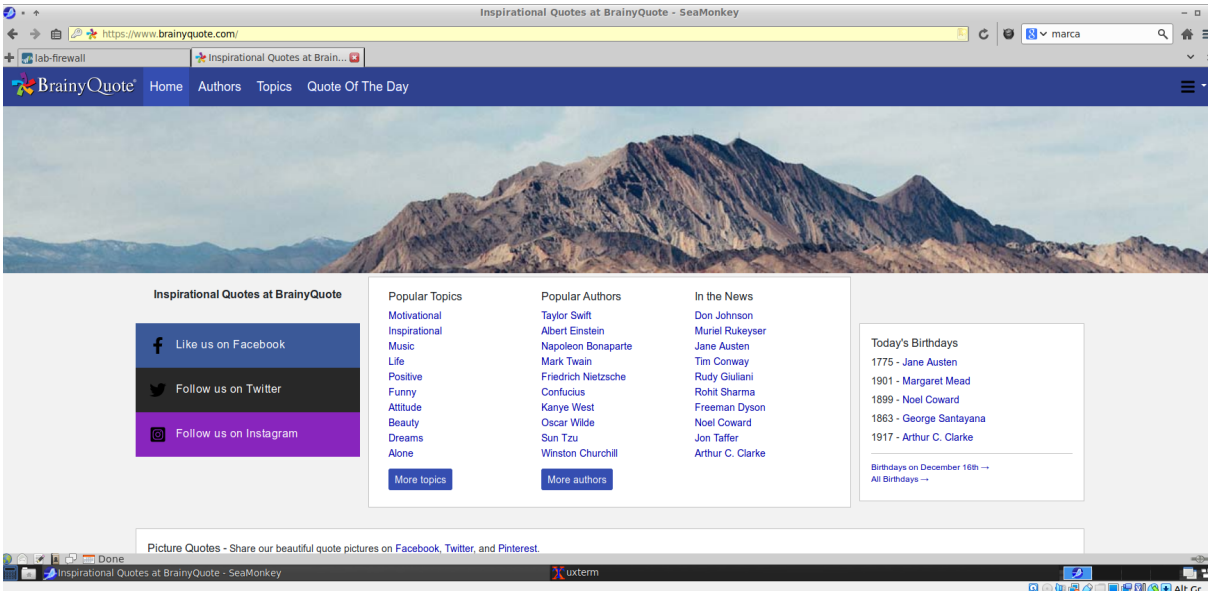

lab-firewall - SeaMonkey
https://192.168.1.254/#device::vsys1::device/user-identification
marca
lab-firewall
paloalto
Dashboard
ACC
Monitor
Policies
Objects
Network
Device
Commit
Config
Search
User Mapping
Connection Security
User-ID Agents
Terminal Services Agents
Group Mapping Settings
Captive Portal Settings
Captive Portal
Enable Captive Portal
Idle Timer (min)
15
Timer (min)
60
GlobalProtect Network Port for Inbound Authentication Prompts (UDP)
4501
Mode
Transparent
Redirect
SSL/TLS Service Profile
None
Authentication Profile
usersDDBB
Certificate Authentication
Certificate Profile
None
NTLM Authentication
Attempts
1
Timeout (sec)
2
Reversion Time (sec)
300
OK
Cancel
admin
Logout
Tasks
Done
lab-firewall - SeaMonkey
uxterm



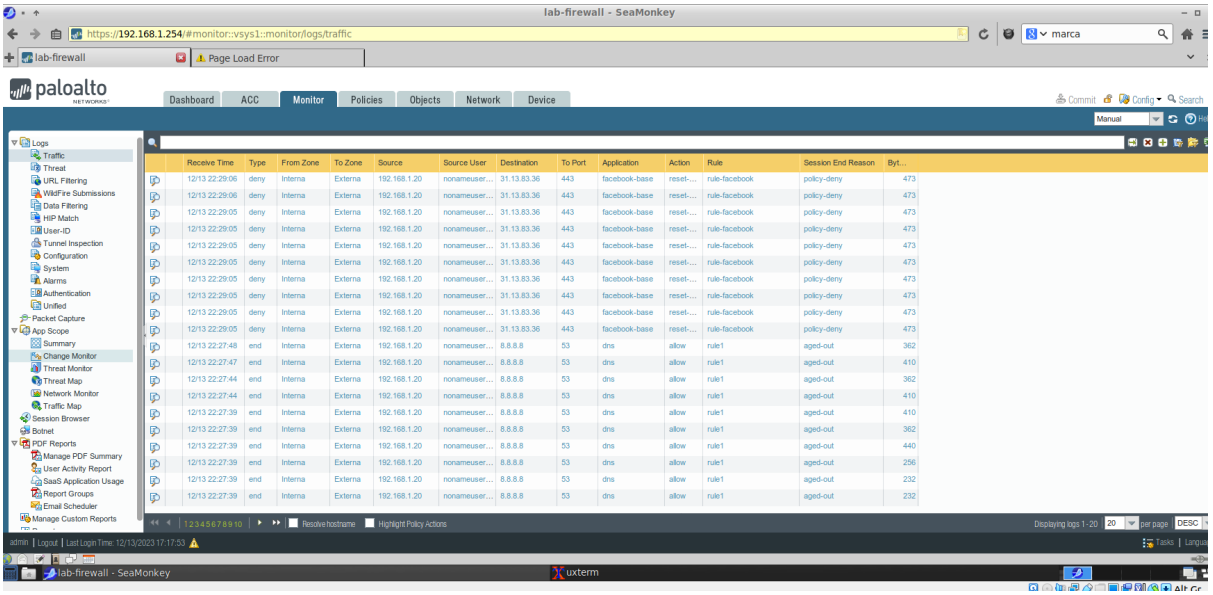

# P15

Al acceder a la página, lo primero que se solicita son las credenciales de acceso que hemos configurado en el usuario anterior. Una vez nos autenticamos podemos acceder a la página.

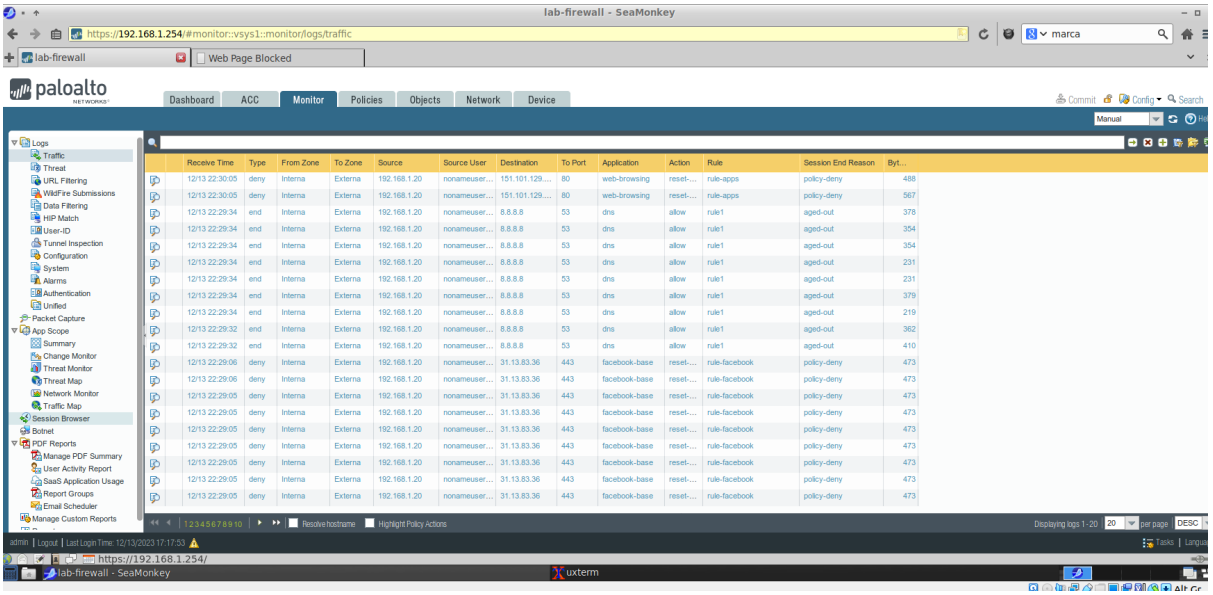


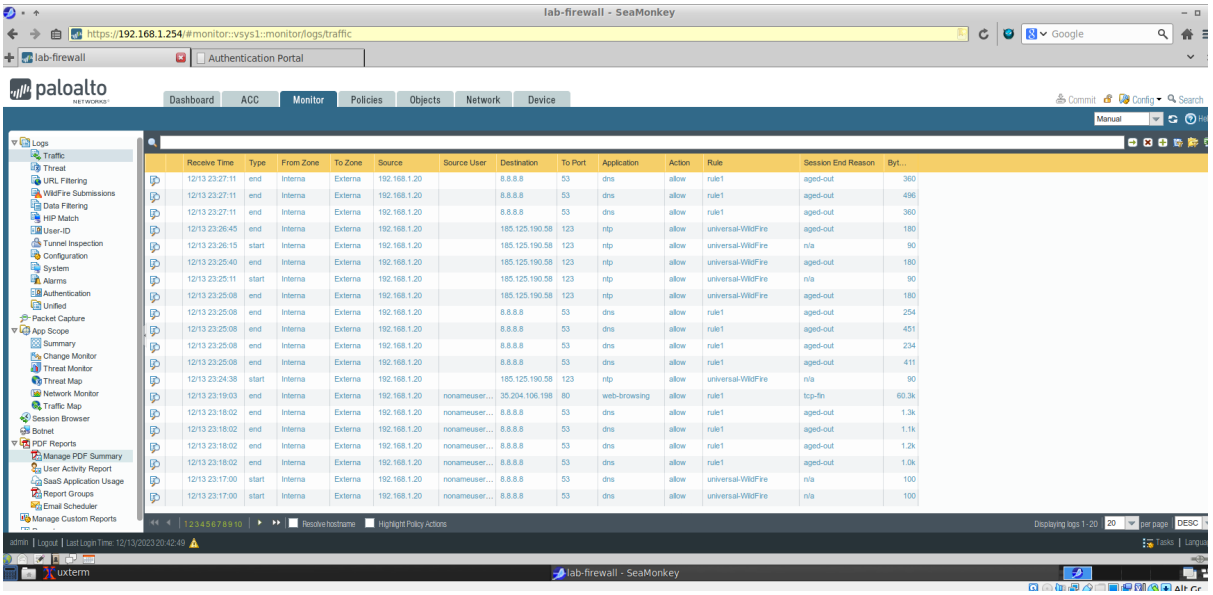
# P16


lab-firewall - SeaMonkey
https://192.168.1.254/#monitor::vsys1::monitor/logs/traffic
Page Load Error
paloalto
Dashboard
ACC
Monitor
Policies
Objects
Network
Device
Traffic
Receive Time
Type
From Zone
To Zone
Source
Source User
Destination
To Port
Application
Action
Rule
Session End Reason
lab-firewall - SeaMonkey
uxterm

# P17

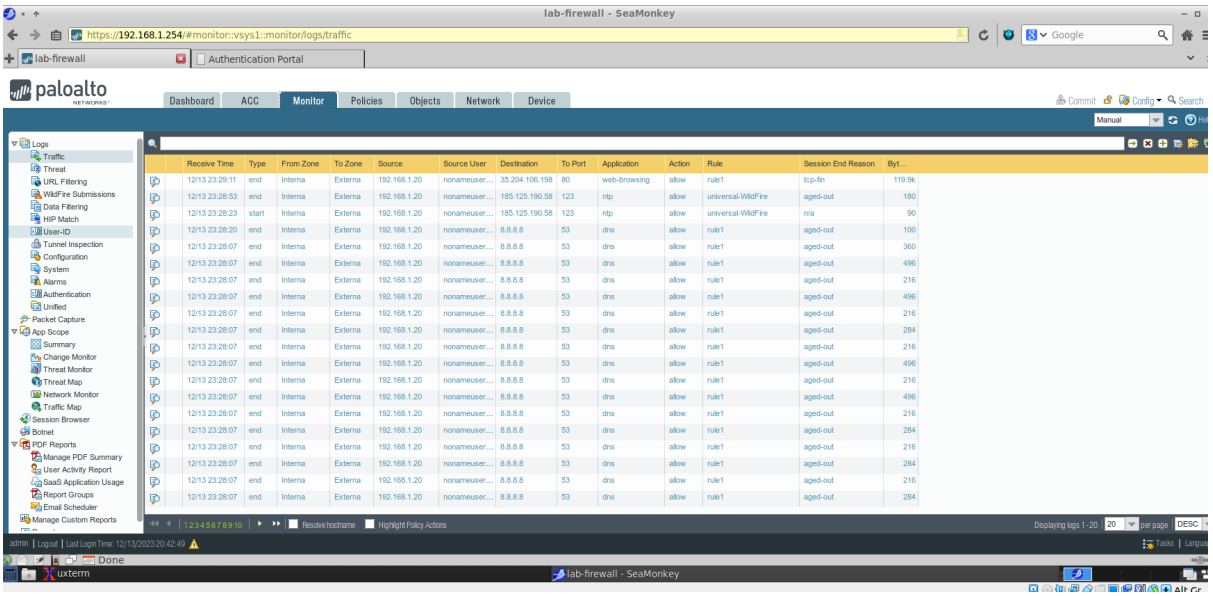

lab-firewall - SeaMonkey
https://192.168.1.254/#monitor::vsys1::monitor/logs/traffic
Web Page Blocked
paloalto
Dashboard
ACC
Monitor
Policies
Objects
Network
Device
Traffic
Receive Time
Type
From Zone
To Zone
Source
Source User
Destination
To Port
Application
Action
Rule
Session End Reason
https://192.168.1.254/
lab-firewall - SeaMonkey
uxterm

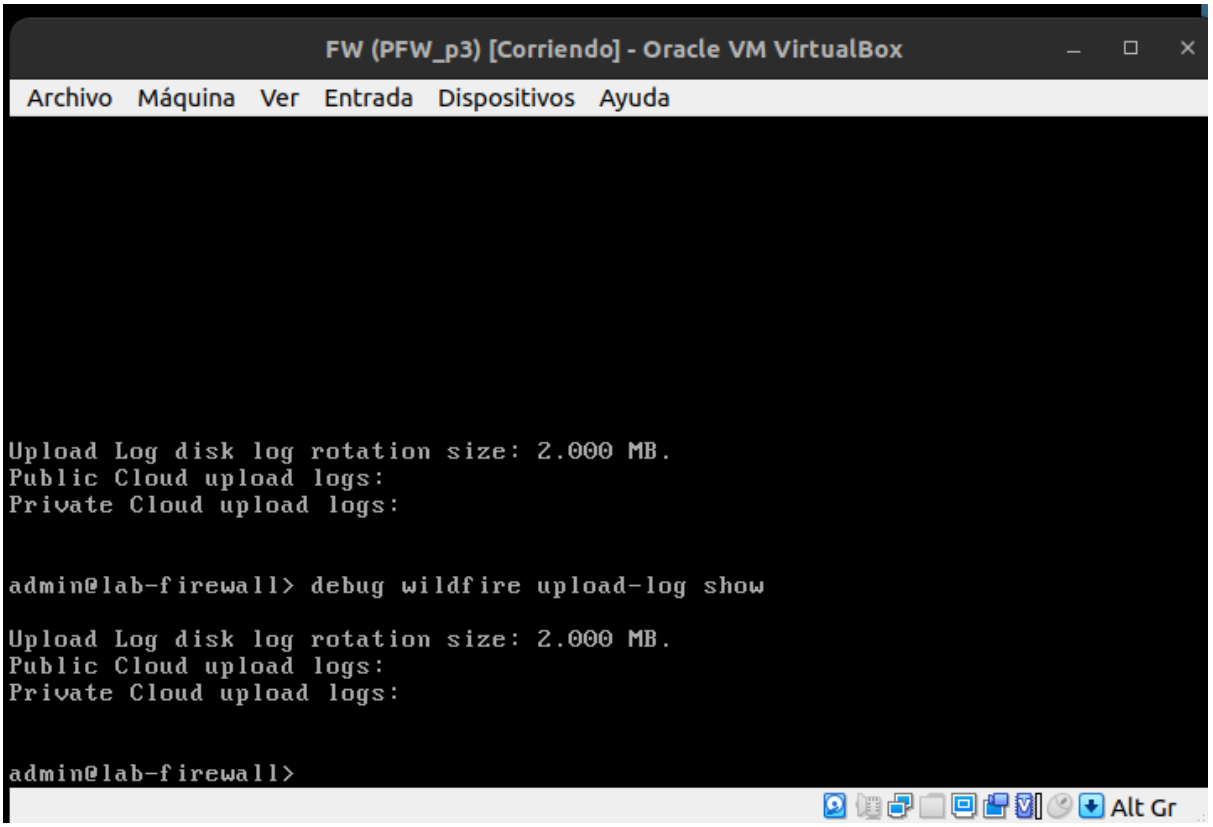
# P18



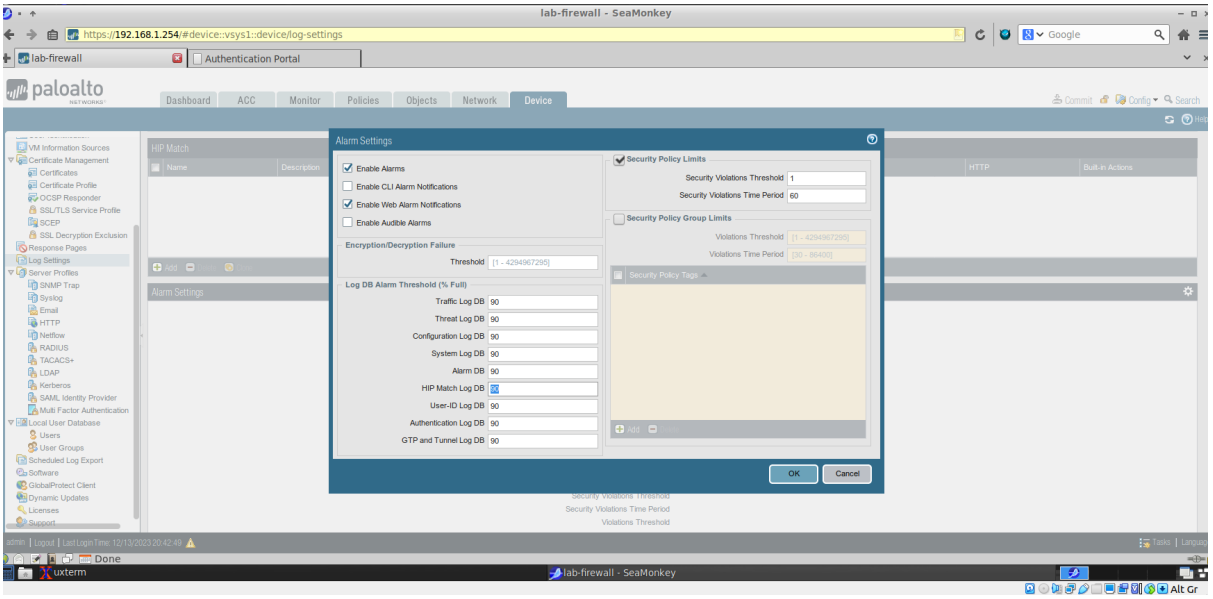

# P19

Aparecen los logs en “Monitor” pero por aunque esperamos más de 5 minutos no aparecen los resultados del análisis.



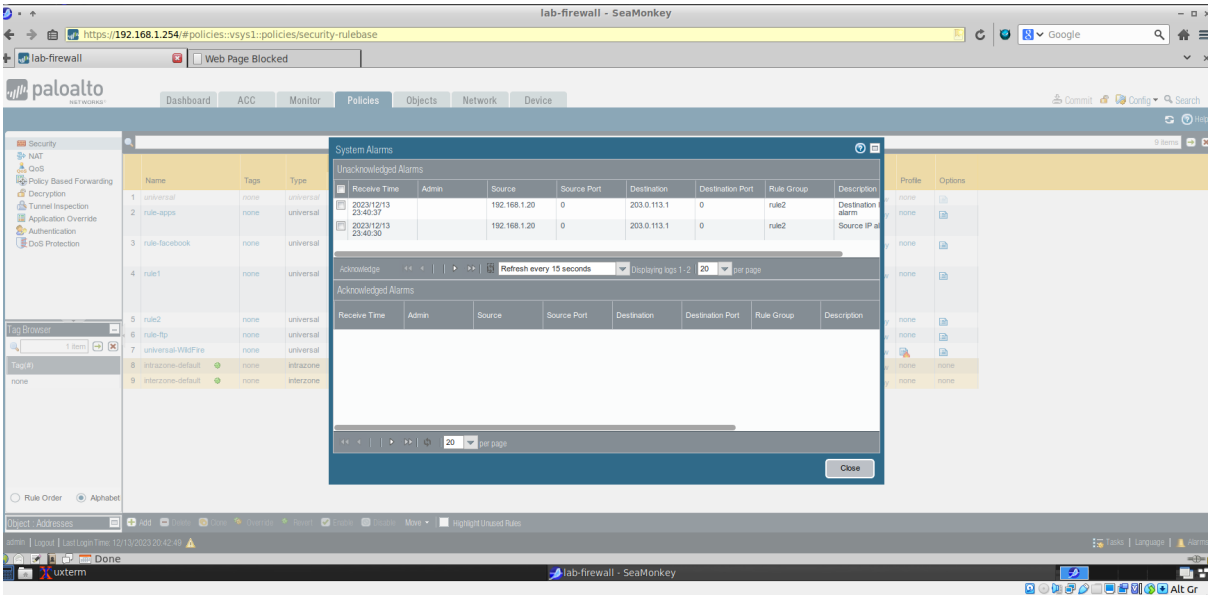



# P20

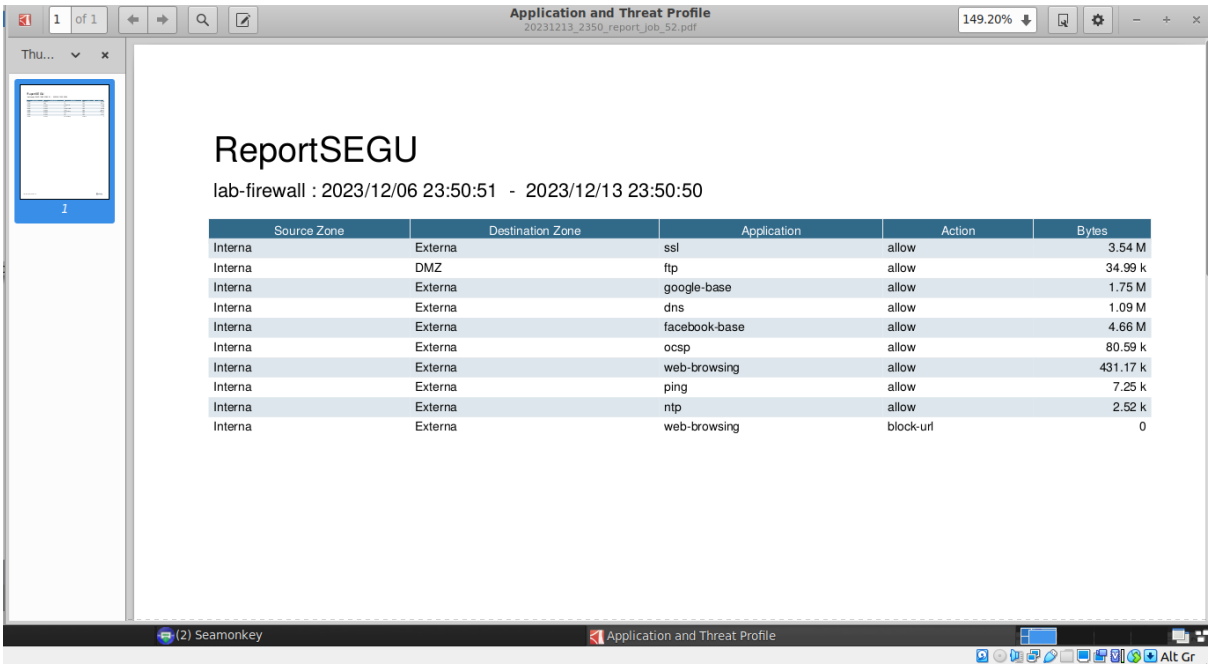



**¿A qué se refiere cada umbral configurado? ¿Cuándo saltará una alarma?**
El umbral “Security Violations Threshold” indica el número mínimo de violaciones de seguridad necesarias para que la alarma se active. Y “Security Violations Time Period” hace referencia al periodo de tiempo mínimo que tiene que transcurrir entre violaciones para que se vuelva a activar la alarma.

# P21




# P22

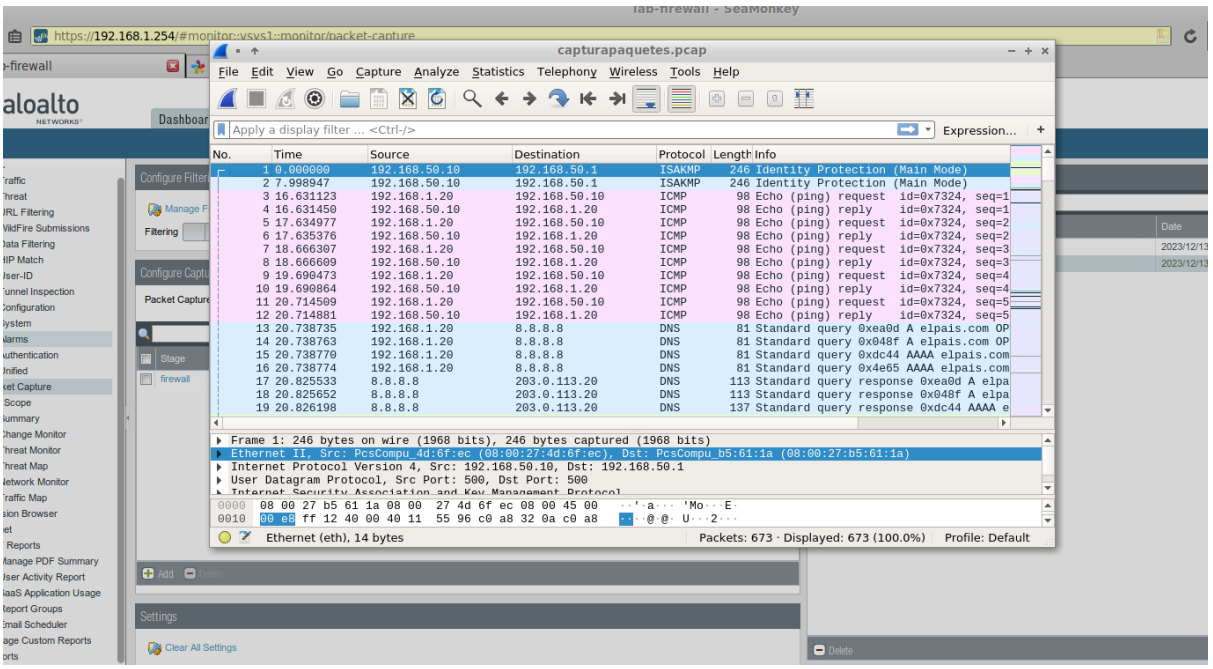



| Source Zone | Destination Zone | Application | Action | Bytes |
|---|---|---|---|---|
| Interna | Externa | ssl | allow | 3.54 M |
| Interna | DMZ | ftp | allow | 34.99 k |
| Interna | Externa | google-base | allow | 1.75 M |
| Interna | Externa | dns | allow | 1.09 M |
| Interna | Externa | facebook-base | allow | 4.66 M |
| Interna | Externa | ocsp | allow | 80.59 k |
| Interna | Externa | web-browsing | allow | 431.17 k |
| Interna | Externa | ping | allow | 7.25 k |
| Interna | Externa | ntp | allow | 2.52 k |
| Interna | Externa | web-browsing | block-url | 0 |

# P23

**¿Qué contiene el fichero?**

El fichero contiene una captura de wireshark en este caso con paquetes ICMP y solicitudes DNS ya que el tráfico de prueba que hemos cursado ha sido un ping al servidor DMZ y una búsqueda web en https://www.brainyquote.com/.


capturapaquetes.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
Expression...
No. Time Source Destination Protocol Length Info
1 0.000000 192.168.50.10 192.168.50.1 ISAKMP 246 Identity Protection (Main Mode)
2 7.998947 192.168.50.10 192.168.50.1 ISAKMP 246 Identity Protection (Main Mode)
3 16.631123 192.168.1.20 192.168.50.10 ICMP 98 Echo (ping) request id=0x7324, seq=1
4 16.631450 192.168.50.10 192.168.1.20 ICMP 98 Echo (ping) reply id=0x7324, seq=1
5 17.634977 192.168.1.20 192.168.50.10 ICMP 98 Echo (ping) request id=0x7324, seq=2
6 17.635376 192.168.50.10 192.168.1.20 ICMP 98 Echo (ping) reply id=0x7324, seq=2
7 18.666307 192.168.1.20 192.168.50.10 ICMP 98 Echo (ping) request id=0x7324, seq=3
8 18.666609 192.168.50.10 192.168.1.20 ICMP 98 Echo (ping) reply id=0x7324, seq=3
9 19.690473 192.168.1.20 192.168.50.10 ICMP 98 Echo (ping) request id=0x7324, seq=4
10 19.690864 192.168.50.10 192.168.1.20 ICMP 98 Echo (ping) reply id=0x7324, seq=4
11 20.714509 192.168.1.20 192.168.50.10 ICMP 98 Echo (ping) request id=0x7324, seq=5
12 20.714881 192.168.50.10 192.168.1.20 ICMP 98 Echo (ping) reply id=0x7324, seq=5
13 20.738735 192.168.1.20 8.8.8.8 DNS 81 Standard query 0xea0d A elpais.com OP
14 20.738763 192.168.1.20 8.8.8.8 DNS 81 Standard query 0x048f A elpais.com OP
15 20.738770 192.168.1.20 8.8.8.8 DNS 81 Standard query 0xdc44 AAAA elpais.com
16 20.738774 192.168.1.20 8.8.8.8 DNS 81 Standard query 0x4e65 AAAA elpais.com
17 20.825533 8.8.8.8 203.0.113.20 DNS 113 Standard query response 0xea0d A elpa
18 20.825652 8.8.8.8 203.0.113.20 DNS 113 Standard query response 0x048f A elpa
19 20.826198 8.8.8.8 203.0.113.20 DNS 137 Standard query response 0xdc44 AAAA e
Frame 1: 246 bytes on wire (1968 bits), 246 bytes captured (1968 bits)
Ethernet II, Src: PcsCompu_4d:6f:ec (08:00:27:4d:6f:ec), Dst: PcsCompu_b5:61:1a (08:00:27:b5:61:1a)
Internet Protocol Version 4, Src: 192.168.50.10, Dst: 192.168.50.1
User Datagram Protocol, Src Port: 500, Dst Port: 500
0000 08 00 27 b5 61 1a 08 00 27 4d 6f ec 08 00 45 00
0010 00 e8 ff 12 40 00 40 11 55 96 c0 a8 32 0a c0 a8
Ethernet (eth), 14 bytes
Packets: 673 · Displayed: 673 (100.0%)
Profile: Default